

;

Diffusion Processes for Inference of Structural Causal Models

IA pour la Science

May 31th, 2026

Inference of Structural Causal Models

The goal is to infer an SCM-model f_G with optimal

- 1) complexity $\mathcal{L}_C$,
- 2) accuracy $\mathcal{L}_D$,
- 3) robustness $\mathcal{L}_{ci}$.

The models:

- 1) the model f_{ai} maps a dataset $\mathbf{D}$ to a model $f_G(\mathbf{w}, \mathbf{x})$,
- 2) the model f_G maps a sample $\mathbf{x} \in \mathbf{D}$ to its forecast $\hat{\mathbf{x}}$, provided optimal parameters $\hat{\mathbf{w}}$.

The graph generation and rewriting

A graph $G(V, E)$ with the

- 1) coloring function $c(V)$,
- 2) weights $\mathbf{w}(V)$,
- 3) probability of the vertices $p(V)$,
- 4) probability of the edges $\mathbf{Z} \in \{0, 1\}^{n \times n}$ or $\mathbf{\Gamma} \in [0, 1]^{n \times n}$.

The graph G defines the function $f_G(\mathbf{w}, \mathbf{x}) \rightarrow \hat{\mathbf{x}}$.

There given a graph rewriting rules $G \mapsto G'$ so that each step delivers an admissible graph (admissible means a graph G defines a function f_G).

To produce a sequence of the graphs introduce a vector $\mathbf{z} = \text{vec}(\mathbf{Z})$.

The van der Pol oscillator mathematical model

The forced Van der Pol oscillator:

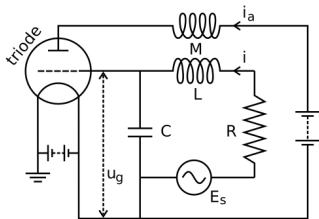
$$\frac{d^2x}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x - A \sin(\omega t) = 0,$$

has two linear and one non-linear parameters: μ, A, ω , where A is the amplitude, of the wave function and ω is its angular velocity.

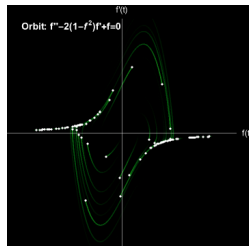
The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 \mapsto \ddot{x}, \dot{x}, x$$

and the time t .

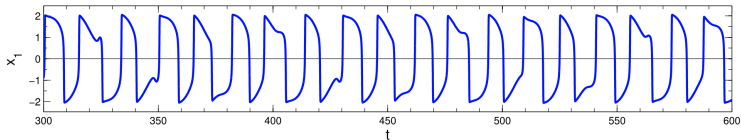


Electrical circuit resulting in a forced Van der Pol oscillator

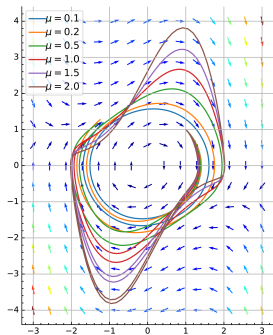


Randomly chosen initial conditions are attracted to a stable orbit

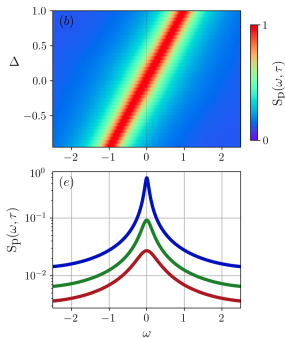
Three spaces to observe the Van der Pol oscillator



Chaotic behaviour in the Van der Pol oscillator with sinusoidal forcing.



Phase portrait of van der Pol's equation



Oscillator power spectra with two parameters

Alternative form of the van der Pol oscillator model

An alternative form is a time-independent Hamiltonian by augmenting it to a four-dimensional, $X_1, \dots, X_4$ autonomous dynamical system using an auxiliary second-order nonlinear differential equation:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \quad \ddot{y} + \mu(1 - x^2)\dot{y} + y = 0.$$

The Hamiltonian

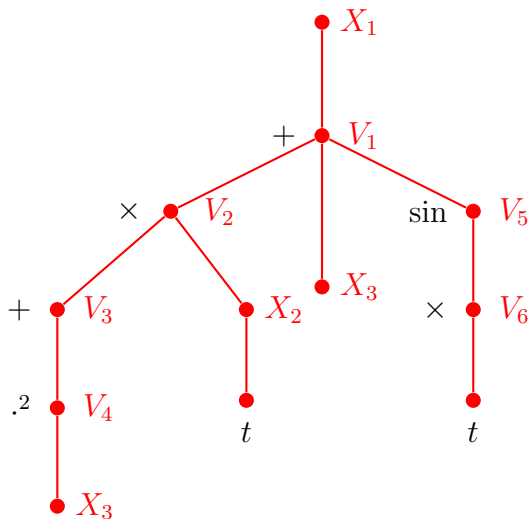
$$H(x, y, p_x, p_y) = p_x p_y + xy - \mu(1 - x^2) y p_y,$$

where

$$p_x = \dot{y} + \mu(1 - x^2)y \quad \text{and} \quad p_y = \dot{x}$$

are the conjugate momenta corresponding to "x" and "y".

The causality graph of Van der Pol oscillator model



The adjacent matrix Z

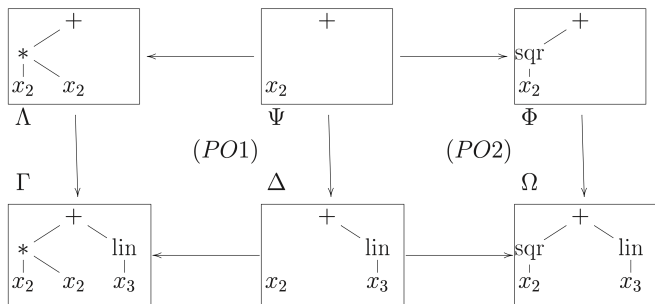
	X_1	X_2	X_3	V_1	V_2	V_3	V_4	V_5	V_6	t
X_1				1						1
X_2										1
X_3										
V_1			1		1			1		
V_2		1				1				
V_3							1			
V_4			1							
V_5									1	
V_6										1

The nodes: $X_1(t)$, $X_2(t)$, X_3 are the observable variables. The latent variables $V_1, \dots, V_6 \in \{\sin(\cdot), (\cdot)^2, +(w_1, \dots, w_{k+1}), \times(\dots, w_0)\}$

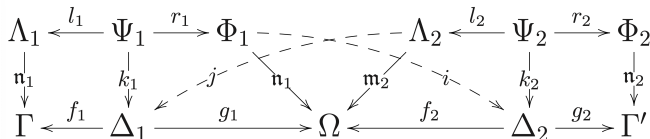
The function «+» has $k + 1$ linear parameters and k arguments.

The function « $\times$ » has one parameters and k' arguments.

Graph rewriting ensures the admissibility of the model f_G ¹

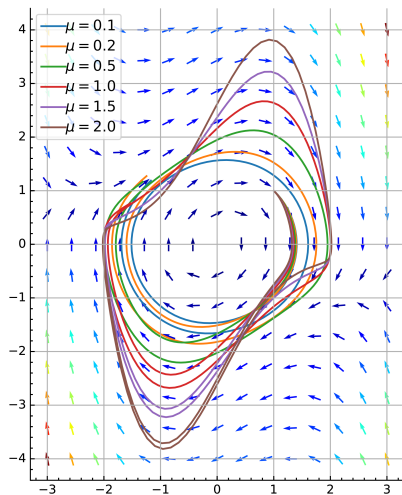


A graph of the form $+(\times(+(\dots)))$ could be simplified, too.



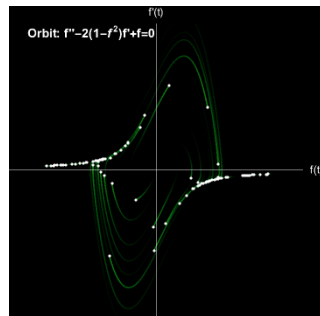
¹This topic will be investigated separately. See Hartmut Ehrig et al. Fundamentals of Algebraic Graph Transformation, Springer, 2006.

Van der Pol oscillator²

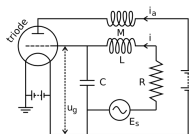


Phase portrait of van der Pol's equation $\frac{d^2 y}{dt^2} + \mu(y^2 - 1) \frac{dy}{dt} + y = 0$.

²Wikipedia: Van der Pol oscillator

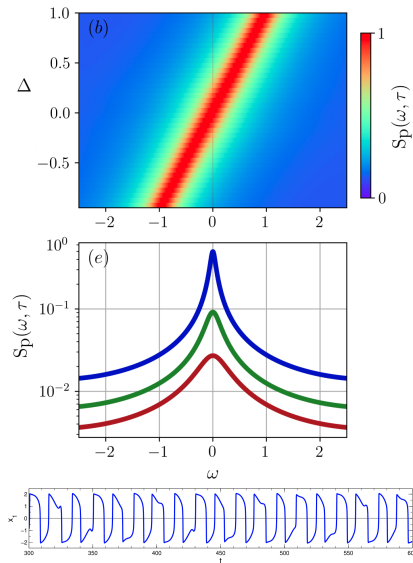


Randomly chosen initial conditions are attracted to a stable orbit



Electrical circuit resulting in a forced Van der Pol oscillator

The van der Pol oscillator model



Chaotic behaviour in the Van der Pol oscillator with sinusoidal forcing. The nonlinear damping parameter is equal to

The van der Pol oscillator model

The forced Van der Pol oscillator:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x - A\sin(\omega t) = 0,$$

has two linear and one non-linear parameters: μ, A, ω , where A is the amplitude, of the wave function and ω is its angular velocity.

The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 = \ddot{x}, \dot{x}, x$$

and the time t .

An alternative form is a time-independent Hamiltonian by augmenting it to a four-dimensional, $X_1, \dots, X_4$ autonomous dynamical system using an auxiliary second-order nonlinear differential equation:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \quad \ddot{y} + \mu(1 - x^2)\dot{y} + y = 0.$$

The Hamiltonian

$$H(x, y, p_x, p_y) = p_x p_y + xy - \mu(1 - x^2) y p_y,$$

where

$$p_x = \dot{y} + \mu(1 - x^2)y \quad \text{and} \quad p_y = \dot{x}$$

are the conjugate momenta corresponding to "x" and "y".

Slide 2

1. Goals
2. Requirements to model
3. How to construct a model
4. Algebraic and Statistical assumptions
5. Do-intervention and testing (test-ability)
6. Local and global modeling expand
7. Graph diffusion and reconstruction
8. Neural ODE is possible
9. Fine-tuning
10. And testing
11. The general probabilistic model and loss function
12. Non-parametric modeling and Optimal sampling
13. Time and space segmentation

The graph probability

Introduce a Metropolis Hastings graph sampling procedure

Models

The goal is to learn to search a model of optimal structure.

The structure must be admissible:

1. An acyclic directed graph with given number of roots and leafs.
2. Its nodes are operations from a given set.
3. It must fit the data and respect statistical assumptions about the data.

Two main approaches are:

1. Symbolic regression is another computationally complex genetic algorithm.
2. Structure learning shows difficulties in delivering admissible structures of models.

The slides below lists variants of problem statement and methods to solve it.